the stretching (we call it the *inflaton field*)[fn3] dumps a massive amount of energy into the cosmos to set up the "hot" phase of the Hot Big Bang. In the original version of the ekpyrotic model, on the other hand, the early universe is heated up by a spectacular collision of two adjacent 3D branes, one of which contains what will later become our whole cosmos. After the collision, the two branes go their separate ways, slowly drifting apart across the bulk and expanding. But they will come back. The ekpyrotic scenario is a cyclic one, with creation and destruction of the cosmos occurring over and over again.

Personally, I find this whole thing makes much more sense if you employ the oldest tool in the physicist's toolbox: hand-waving.

Your left hand is our 3-brane: the 3-dimensional universe in which we live. (Obviously, none of this is to scale. This is hand-waving, after all.) Your right hand is another, "hidden" brane.[fn4] First, place your hands together, with your fingers closed, in a prayer posture. This is the moment of cosmic creation. This is the collision that sets off the primordial fire. Both branes are filled with dense hot plasma at this moment, an unimaginably intense inferno forging the first atoms and carrying the humming plasma waves that, on our brane, we'll later see as fluctuations in the cosmic microwave background light. Now, slowly separate your hands to a short distance apart, keeping them parallel, and spread your fingers. The branes have drifted apart across the higher-dimensional bulk, and the space in each brane is, independently, cooling and expanding in its own way. There's no inflation phase in this model, just a steady expansion after the collision. And they're not expanding into the bulk between them; they're extending in their own branes parallel to each other. On our brane, your left hand, this is the cosmos we see today. While we can't perceive our motion away from the other brane, we do see galaxies recede into the distance as the 3D space we live in expands, and our universe becomes more and more empty, heading toward a Heat Death. We don't know what's happening on your right hand, the hidden brane. Maybe there are civilizations there too, watching their own universe empty out as it traverses an unseen void. Maybe it's a quiet, desolate place, where for whatever reason matter never learned to arrange itself into life. Maybe they have puppies who can talk. Unless we somehow detect a gravitational wave signal from the hidden brane, we may never know its true nature, or if it even exists.